

Customer Churn Prediction & Business Intelligence Report

Automated analysis of customer churn drivers, model performance, and retention recommendations

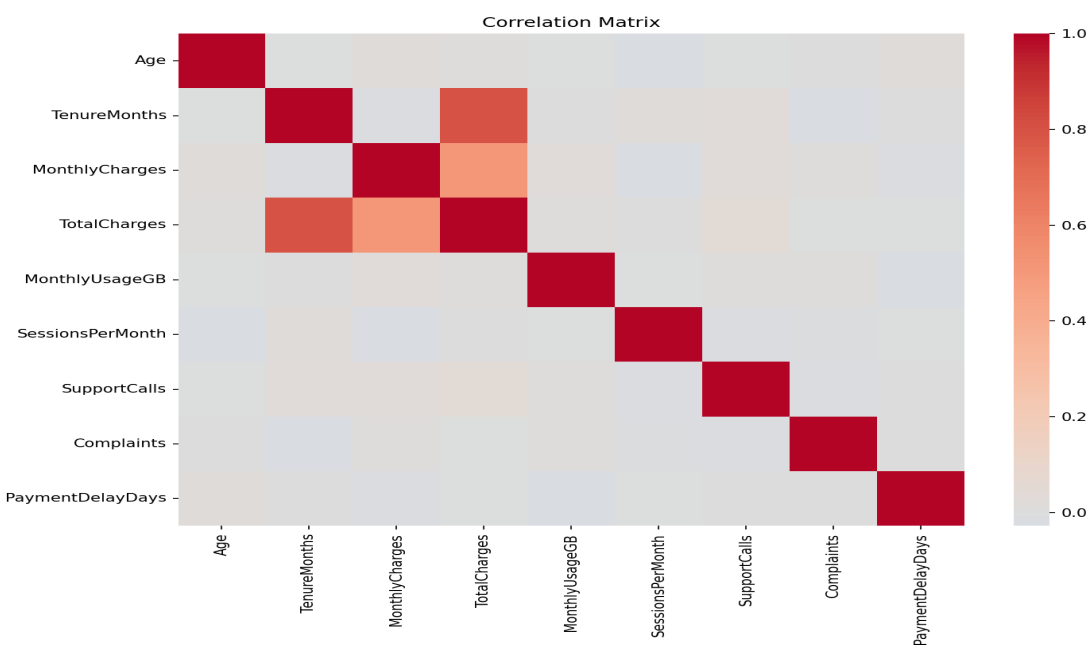
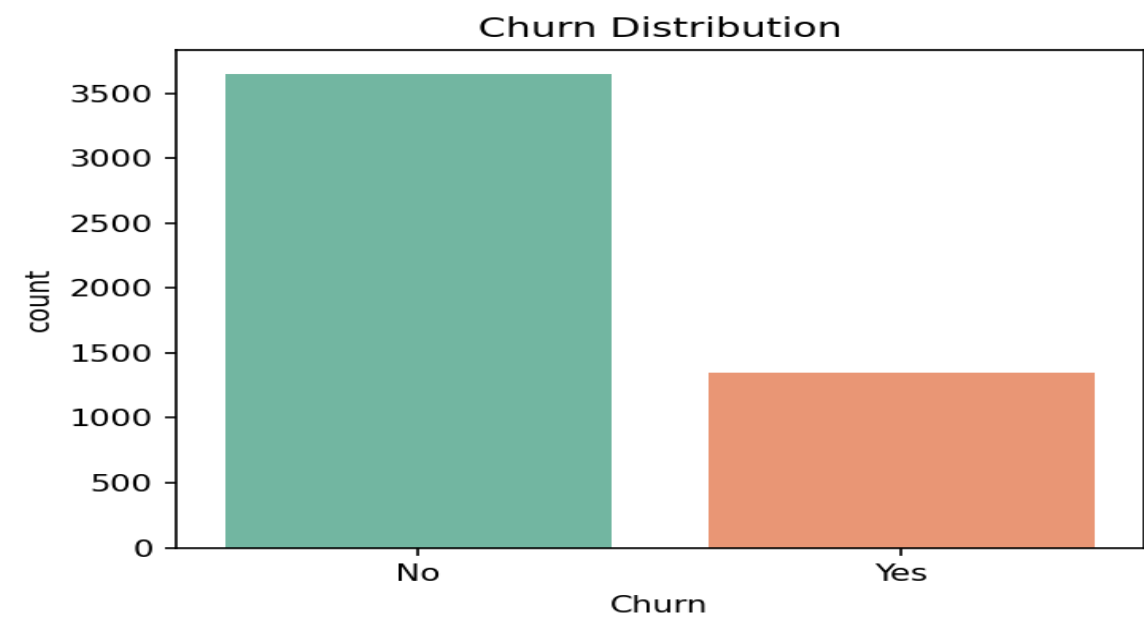
Executive Summary

Total Customers	5,000
Current Churn Rate	27.0%
Average Monthly Charges	\$64.88
Monthly Revenue at Risk	\$91,582.34
Best Performing Model	Gradient Boosting

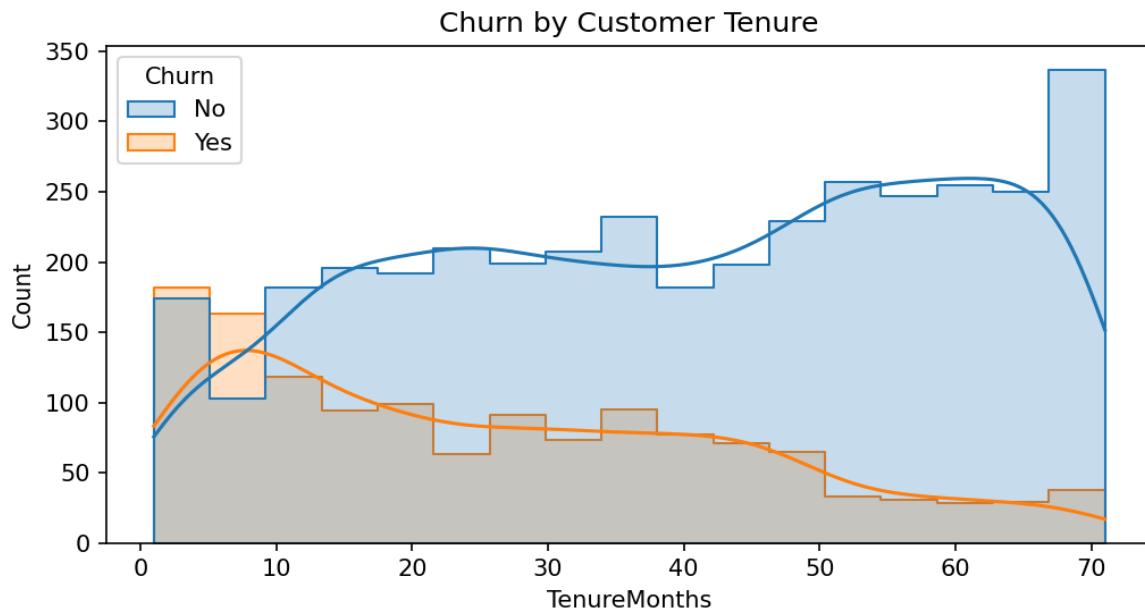
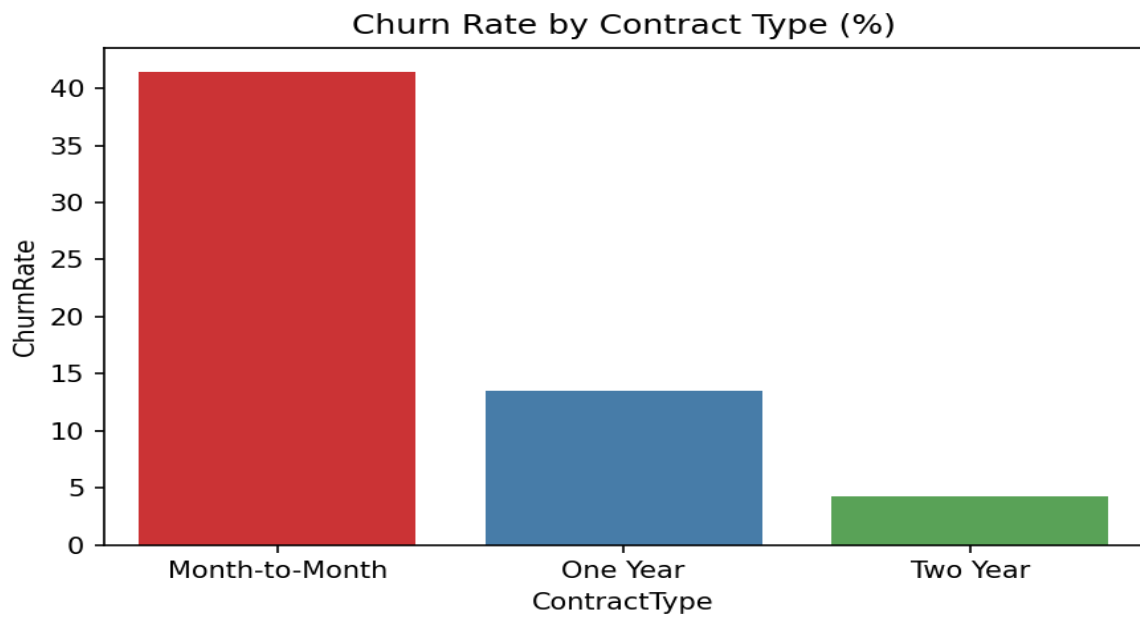
Model Performance Comparison

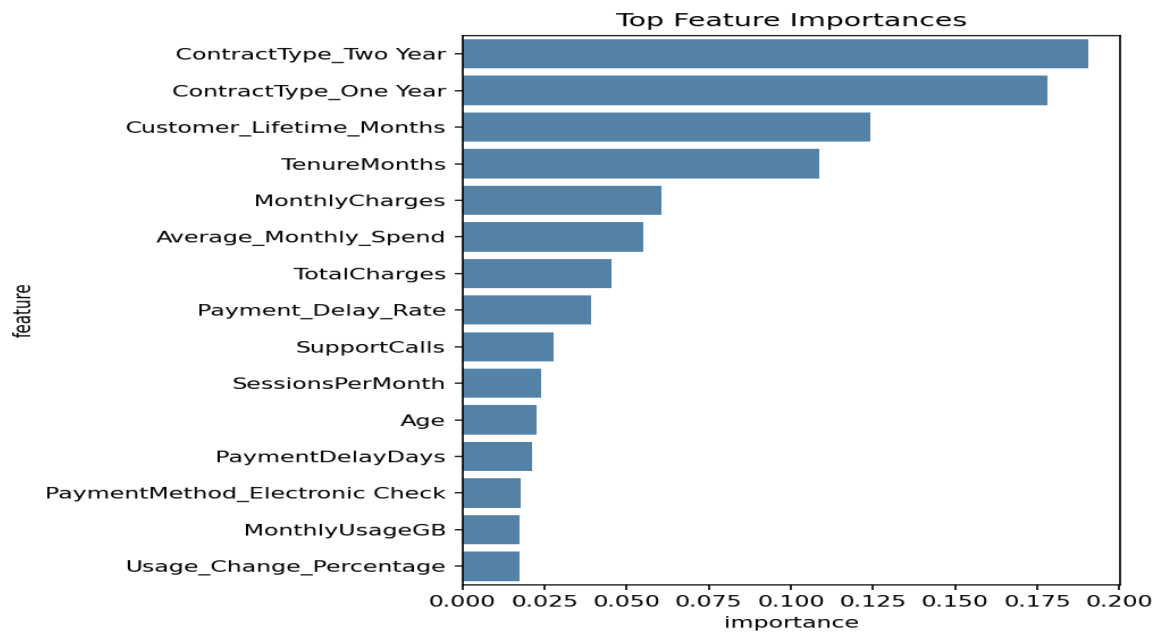
Model	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC
Logistic Regression	0.749	0.524	0.763	0.621	0.817	0.606
Decision Tree	0.716	0.485	0.822	0.610	0.798	0.555
Random Forest	0.760	0.540	0.752	0.628	0.821	0.606
Gradient Boosting	0.794	0.678	0.452	0.542	0.828	0.647
XGBoost	0.797	0.685	0.459	0.550	0.816	0.611

Churn Overview

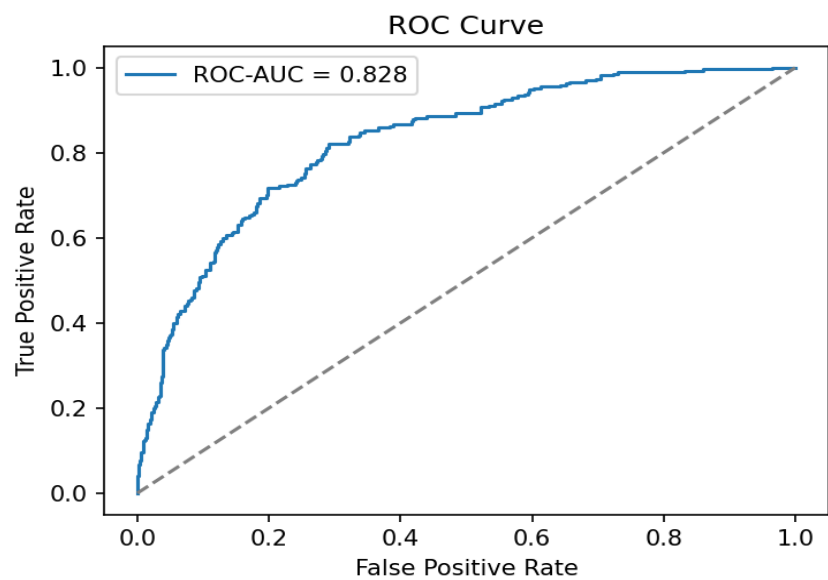
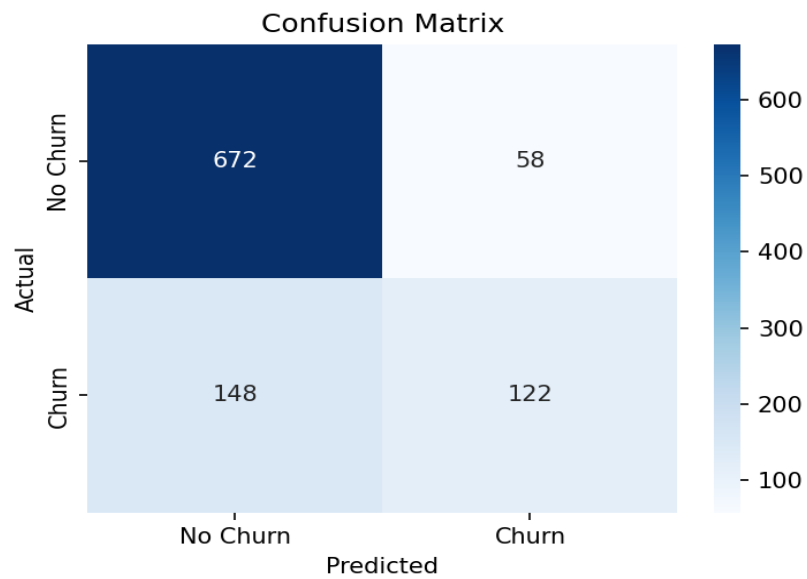


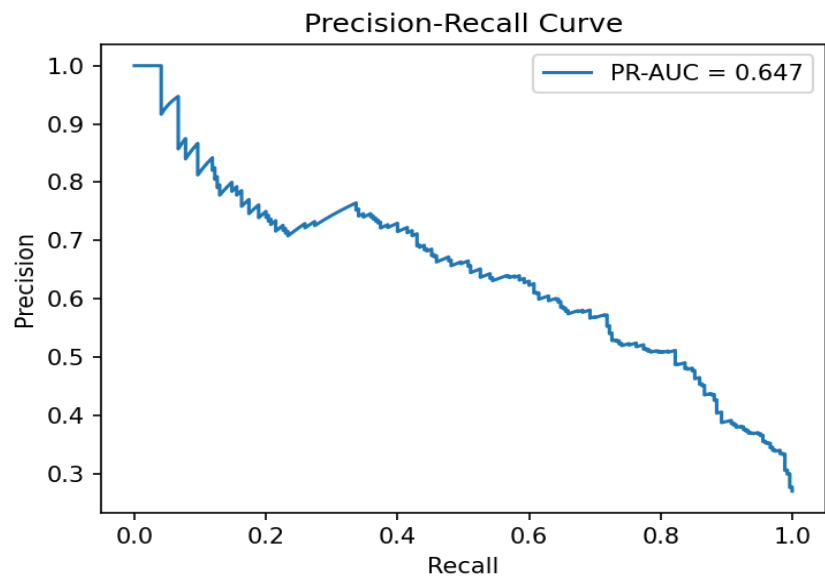
Churn Drivers



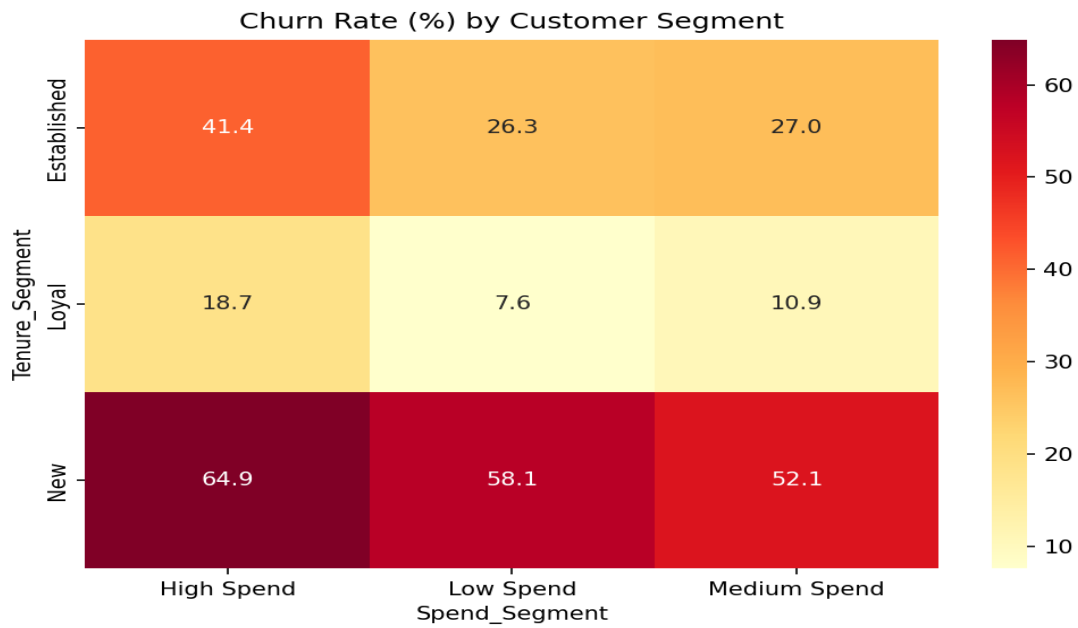


Model Diagnostics





Customer Segmentation



Recommendations

- Prioritize retention outreach for month-to-month contract customers, which show the highest churn rate.
- Offer discounted annual or two-year contract upgrades to high-risk, high-spend customers.
- Investigate service quality for customers with 4 or more support calls, where churn risk rises sharply.
- Review electronic check as a payment method, since it correlates with higher churn and payment delays.
- Deploy the trained model in the Streamlit dashboard for ongoing, real-time churn monitoring.